

DR. ARUNAPRASANNA SIMHAMBHATLA

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Healthcare Data and Research Analyst and Pharm.D. Graduate with 3+ years of experience analyzing medical and pharmacy claims for employer groups across insurance networks. Proficient in SQL, Python, R, dbt, and Redshift for auditing billing patterns, comparing in-vs out-of-network rates, and identifying overspending via price transparency data. Skilled in building dashboards and ETL pipelines to track healthcare utilization, support value-based care models, and drive clinical quality improvement with strong data governance.

Education

Indiana University Indianapolis, United States

Aug 2023 - May 2025

Master of Science in Health Informatics

GPA: 3.9/4.0

Andhra University, India

Aug 2016 - Oct 2022

Doctor of Pharmacy (Pharm.D)

GPA: 3.5/4.0

Technical Skills

- **Programming & Analytics:** Python (pandas, NumPy, seaborn, matplotlib, pyspark), R, SQL (joins, CTEs, filters, aggregations, window functions), REST APIs, SAS.
- **Machine Learning & Statistics:** Logistic Regression, Random Forest, XGBoost, SVM, NLP (preprocessing), Time Series Forecasting, ROC-AUC, Cross-Validation, Feature Selection, Hypothesis Testing (t-tests, ANOVA, chi-square), Correlation Analysis
- **Data Engineering & Storage:** Redshift, Snowflake, PostgreSQL, ClickHouse, MySQL, MongoDB, BigQuery, AWS S3, dbt, Airflow, ETL, Data Parsing (JSON/XML)
- **Data Visualization & BI:** AWS QuickSight, Tableau, Power BI (DAX, dashboards), Excel (VLOOKUP, macros, PivotTables), KPI Dashboards, Report Automation, Ad Hoc Reporting
- **Healthcare IT & Standards:** Epic, Cerner, Athenahealth, REDCap, OpenEMR, HL7, FHIR, OMOP CDM, ICD-10, CPT, HCPCS, SNOMED CT, RxNorm, NDC, LOINC, MS-DRG
- **Compliance & Methodologies:** HIPAA, IRB Protocols, PHI, PII Compliance, Authorization Workflows, QA Testing, Metadata Documentation, Agile, Scrum, Lean Six Sigma, Business Process Review
- **Tools & Collaboration:** Git, VS Code, Postman, Jira, Confluence, Figma, Microsoft Office Suite, Qualtrics, Slack

Professional Experience

Indiana University School of Medicine

Jan 2025 – May 2025

Research Data Analyst - HANDS in Autism (Riley Pediatric Care Center)

Indianapolis, IN

- Analyzed longitudinal REDCap survey data from 500+ participants completing multi-week autism training programs; calculated weighted knowledge scores and tracked learning retention across pre-, post-, and follow-up assessments using Python, SQL, and Excel for outcome evaluation and program impact analysis.
- Conducted statistical analysis to evaluate training effectiveness by participant role, experience, and year; built and maintained SQL queries and ETL workflows to clean, merge, and standardize survey and clinical datasets, enabling reproducible case-based evaluation and identifying a 25-30% average improvement in post-training knowledge retention.
- Designed automated Power BI dashboards integrating Canvas, Qualtrics, and REDCap to visualize participant outcomes, severity trends, and training impact; collaborated with clinicians and educators to define metrics, document metadata, and support IRB-reviewed reporting and manuscript development.

TALON

May 2024 – Aug 2024

Healthcare Data Analyst Intern

Portsmouth, NH

- Queried and audited medical and pharmacy claims across 100+ employer groups using SQL (Redshift, Snowflake, ClickHouse) and MongoDB to detect adjudication errors, tier leakage, and billing outliers, revealing 15–30% overspending on high-cost services.
- Built scalable ETL pipelines (Python, dbt, AWS S3) to join All Payers Claims Database with MRFs & chargemaster data using NPI-CPT-modifier triplets, enabling bundle-level benchmarking, Transparency in Coverage rule, & UAPA-aligned pricing logic.
- Developed a centralized analytics data mart with SDOH, Pharmacy Benefit Management, and population health indicators to support clinical quality improvement and value-based care, identifying \$7–8M in potential annual savings.
- Conducted PHI-compliant QA audits and maintained version-controlled workflows in Git and Confluence; validated adjudication logic, incentive rules, and reshipping triggers across medical and pharmacy claims.
- Delivered dashboards and cost-saving insights for TALON's MyMedicalMetrics™ platform using QuickSight, Tableau, and Excel; visualized overspend trends, utilization gaps, and network optimization strategies.

Indiana University School of Dentistry

Jan 2024 – May 2025

Research Data Analyst – XRCPEL

Indianapolis, IN

- Conducted REDCap-based usability studies with 100+ dental students on XR device feasibility and empathy training; analyzed surveys using Python, Excel, and Qualtrics with optimized queries for large datasets.
- Integrated EMR and REDCap datasets for oncology and special needs patients undergoing dental procedures; applied data warehousing concepts to perform IRB-compliant data analysis evaluating XR impact on pain perception and care experience.
- Developed Tableau dashboards and authored SOPs and protocol enhancements to support NIH-style reporting, metadata tracking, and institutional presentations; mentoring emerging leaders on dashboard usage and interpretation.

Indiana University Indianapolis

Jan 2024 – May 2025

Graduate Teaching Assistant (INFO - B 505 Informatics Project Management)

Indianapolis, IN

- Mentored 150+ graduate students in Agile, Waterfall, and hybrid methodologies; coached sprint planning, backlog grooming, and risk mitigation for healthcare informatics capstone projects.

- Advised 25+ student teams on designing analytics workflows, defining stakeholder requirements, and aligning deliverables with real-world project timelines; supported documentation and QA reviews.

Indiana University School of Liberal Arts

Aug 2023 – Jan 2024

Lead Research Director – Virtual Reality

Indianapolis, IN

- Led cross-functional development of a VR-based behavioral intervention targeting public speaking anxiety; deployed to 500+ students with screening logic and survey-based engagement tracking.
- Conducted outcome analysis using Python and Excel; generated population-level behavioral metrics and ad hoc dashboards to quantify confidence improvement and support program refinement.

Rajiv Gandhi Institute of Medical Sciences

Sep 2022 - Jun 2023

Clinical Research Coordinator

India

- Coordinated multi-site Phase II/III oncology trials, overseeing EMR-integrated data collection, eConsent workflows, and GCP-aligned documentation for 100+ participants; ensured protocol adherence from site activation to closeout.
- Performed eligibility screening, AE tracking, and CRF data entry in REDCap and EHR systems; collaborated with investigators and data managers to resolve queries, support interim analyses, and maintain audit-ready datasets for sponsor reviews.

Rajiv Gandhi Institute of Medical Sciences

Jul 2021 - Oct 2022

Clinical Pharmacist Intern

India

- Documented ICD-10 diagnoses and RxNorm-coded medications in EMRs to support prescribing accuracy, medication reconciliation, and formulary compliance across inpatient settings.
- Collaborated on daily ward rounds to optimize pharmacotherapy; tracked adverse drug reactions (ADRs), validated therapy changes, and supported pharmacovigilance reporting in alignment with quality and safety standards.

Applied Informatics and ML Projects

Pathology Data Curation & EDA (TCGA-GBM)

March 2025 - May 2025

- Curated a simulated TCGA-GBM pathology dataset by parsing and normalizing deeply nested JSON structures using Python; transformed raw records into structured tables containing tumor stage, morphology, Ki-67 index, and mitotic rate to support oncology ML workflows.
- Simulated missing data patterns and applied imputation strategies to enhance dataset robustness; performed exploratory data analysis using pandas, seaborn, and matplotlib to identify biomarker distribution shifts across tumor grades and patient cohorts.
- Designed visual dashboards to interpret feature relationships, detect outliers, and support downstream feature engineering and hypothesis testing for predictive modeling.

CDSS for Asthma and Colorectal Cancer (*OpenEMR, OHDSI ATLAS, HL7 FHIR, Python*)

Aug 2024 - Dec 2024

- Designed and implemented rule-based Clinical Decision Rules (CDRs) in OpenEMR to enhance asthma diagnosis (FeNO, spirometry) and biomarker-driven treatment for colorectal cancer (KRAS/BRAF).
- Integrated patient cohorts from OHDSI ATLAS and simulated real-time HL7 FHIR-based CDSS responses using REST APIs; improved guideline adherence, diagnostic precision, and treatment consistency.
- Enhanced provider usability by incorporating explainability logic, standardized clinical pathways, and decision audit trails into the CDSS workflow.

Unified Clinical Data Integration System for HIE Interoperability (*NLP, Python, JSON*)

Aug 2024 - Dec 2024

- Developed an AI-powered system to unify structured and unstructured EHR data across Health Information Exchanges (HIEs) using NLP and HL7 FHIR, improving longitudinal patient visibility.
- Validated the platform in oncology use cases, demonstrating 25% faster diagnostic decisions and projecting \$70M+ in 3-year healthcare cost savings through duplicate test reduction and care coordination.

Breast Cancer Risk Prediction

Jan 2024 - May 2024

- Performed exploratory data analysis and statistical hypothesis testing (Mann-Whitney U, Chi-square) on 30 tumor features from the WDBC dataset to identify key markers associated with malignancy.
- Built and evaluated a logistic regression model in R, achieving 99.16% classification accuracy; visualized model diagnostics using ROC curves, boxplots, and heatmaps to support early breast cancer detection and feature interpretability.

EHR Interoperability via XML Schema

Jan 2024 - May 2024

- Designed a FHIR-compliant XML schema with datatype constraints, validation logic, and field-level rules to standardize EHR data exchange across systems.
- Mapped diagnosis, medication, and procedure fields to global standards (SNOMED CT, RxNorm, CPT) using XSLT and Postman APIs, resolving structural mismatches and improving semantic consistency.

PCOD Diagnosis using Machine Learning

Aug 2023 - Dec 2023

- Built and evaluated machine learning models (XGBoost, Random Forest, Decision Tree, Logistic Regression) for PCOD classification using hospital diagnostic data; achieved 92% accuracy and 91% AUC-ROC with Repeated Stratified K-Fold cross-validation.
- Automated data pipelines for preprocessing (scaling, encoding, outlier detection), conducted EDA in Python and Snowflake, and identified key predictors through feature importance analysis to support clinical decision-making and algorithmic transparency.

Achievements & Certifications

- Six Sigma Green Belt Certified - LinkedIn Learning ; Lean Six Sigma White Belt Learner - CertiProf
- HIPAA Awareness for Business Associates - HIPAA Training
- Board Certified Clinical Pharmacist, certified by the Pharmacy Council of India.

Publications

- Simhambhatla, A.P., Dormer, D., Gali, N.C., Kaja, L., Darla, M., et al. *Students consider the impact of Democratic and Republican Healthcare Policies on Health Informatics and Health Information Technology*. Preprint, December 2024. **ResearchGate**